Interactive Hand Gesture Recognition for Children's Early Educational Tools

A Feasible, Impactful Computer Vision Project for Community & Educational Use

Project Phase: Milestone 1 — Phase 1 of 5 (CP1)	**Target Submission:** Thu 15 Oct, 11:30 PM
In-Class Pitch: Friday 16 Oct (3-Minute Demo)	**Target Git Tag:** cp1-proposal
Scope / Team: Solo Project / Small Community Tool	**Repository Status:** Public / GitHub Stub Ready

1. Project Title & Overview

System Title: Real-Time Educational Hand Gesture Recognition for Interactive Early Childhood Learning

This project establishes an accessible, real-time computer vision system capable of recognizing static hand gestures (e.g., numbers 0–5, thumbs up/down, peace sign) captured via standard webcam or mobile phone camera. Designed specifically for a small community classroom / children's learning center, the system turns physical gesture inputs into interactive feedback for digital learning games without requiring specialized hardware.

2. Problem Statement

Early childhood education benefits greatly from tactile and gesture-based interaction, yet community learning centers and under-resourced classrooms rarely have access to expensive touchscreen devices or specialized sensor hardware (such as Kinect or Leap Motion). Existing software interfaces rely heavily on traditional mouse and keyboard navigation, which can be challenging for young children developing fine motor skills.

A software-based computer vision approach solves this barrier by using existing commodity webcams or low-cost camera modules. By transforming simple hand gestures into instant educational triggers (such as answering counting quizzes or selecting interactive items), vision technology delivers an engaging, hands-free interface at zero additional hardware cost.

3. Objectives and Measurable Outcomes

The primary goal is to build an end-to-end vision pipeline capable of detecting and classifying 6 hand gesture classes from live or recorded video frames.

- **Gesture Recognition Accuracy:** Achieve a macro-F1 score of at least 0.85 across 6 static gesture classes on a held-out test set under varied lighting conditions.
- **Real-Time Inference Speed:** Maintain live frame processing at ≥ 15 FPS on standard laptop hardware (CPU only).
- **Data Evidence & Feasibility:** Assemble a dataset of 120+ labeled gesture images (20+ per class) with a 20-sample contact sheet and verified feasibility spike notebook.
- **Minimum Viable Fallback (Safety Net):** Deliver a baseline skin-color segmentation and contour-convexity classifier operating at ≥ 20 FPS with macro-F1 ≥ 0.70 if deep learning fine-tuning falls behind schedule.

4. Proposed Computer Vision Approach & Algorithms

The system is designed with a clear two-tier methodology: a fast classical baseline that acts as a safety net, and a robust main method using lightweight deep learning.

Pipeline Stages:

- **1. Acquisition & Preprocessing:** Webcam frame capture (640x480), Gaussian smoothing, and RGB-to-HSV color space conversion.
- **2. Baseline Method (Classical Vision):** YCbCr / HSV skin color thresholding, morphological opening/closing, followed by contour extraction and Convexity Defects analysis paired with a k-Nearest Neighbors (kNN) or SVM classifier.
- **3. Main Method (Deep Learning / Transfer Learning):** Hand region ROI extraction, landmark detection via MediaPipe / fine-tuned MobileNetV2 architecture trained on gesture embeddings.
- **4. Post-Processing & Output:** Smoothing predictions across a 3-frame rolling window to eliminate jitter, displaying bounding boxes, landmark overlays, and class labels with confidence scores.

5. System Architecture & Workflow Diagram



Figure 1: End-to-end Computer Vision Pipeline architecture showing baseline and main classification paths.

6. Dataset Specification & Data Evidence

Primary Data Source: Public HaGRID (Hand Gesture Recognition Image Dataset) combined with custom captured phone camera samples under cluttered/classroom lighting.

Parameter	Specification
Total Sample Count	150 labeled image samples (25 samples / class)
Gesture Classes (6)	0 (Fist), 1 (One/Index), 2 (Peace/V), 5 (Open Palm), Thumbs Up, Thumbs Down
Conditions Covered	Plain white background, cluttered classroom desk, varied indoor warm/cool lighting
Data Split	70% Training (105 imgs), 15% Validation (22 imgs), 15% Test (23 imgs)

Licensing & Evidence	CC BY-NC 4.0 / Public Domain. 20-sample contact sheet committed under data/samples/

7. Task Assignments & Module Ownership

Module / Task	Owner	Deliverables
Data Collection & Contact Sheet	Lead Developer	Dataset curation, 20-sample grid, labels JSON
Feasibility Spike & Baseline	Lead Developer	Colab notebook, Canny/HSV skin mask script
Main Method (MobileNet/MediaPipe)	Lead Developer	Model fine-tuning, weight checkpoints, evaluation script
Pipeline Integration & Repo	Lead Developer	Real-time webcam inference, README, git tags

8. Project Timeline Aligned with 5 Checkpoints

Checkpoint / Phase	Hand-In Deadline	In-Class Check	Points
CP1: Proposal & Feasibility Spike	Thu 15 Oct, 11:30 PM	Fri 16 Oct	10 pts
CP2: 50% Progress (Baseline Working)	Thu 22 Oct, 11:30 PM	Fri 23 Oct	15 pts
CP3: Core Implementation & Main Model	Thu 29 Oct, 11:30 PM	Fri 30 Oct	15 pts
CP4: 80% Progress & Evaluation	Thu 12 Nov, 11:30 PM	Fri 13 Nov	20 pts
CP5: Final Presentation & Live Demo	Thu 19 Nov, 11:30 PM	Fri 20 Nov	30 pts
CP6: Technical Defense	Thu 19 Nov, 11:30 PM	Team Slot	10 pts

9. Risks & Fallback Management

Identified Risk	Impact / Likelihood	Mitigation & Fallback Strategy
Poor Segmentation in Cluttered Lighting	High / Medium	Fallback to MediaPipe hand landmark detection which relies on bounding box proposals rather than naive color skin thresholding.
Deep Model Training Latency on CPU	Medium / High	Use lightweight MobileNetV2 features or quantized ONNX model to maintain ≥15 FPS live demo capability.